

CLAF-Net: A Cross-Lingual Asymmetric Focal Network for Zero-Shot Disaster Response Classification

1st Farah Hamna M

*Dept of Information Technology
Karpagam College of Engineering
Coimbatore, India
farahhamna1608@gmail.com*

2nd Rishikesh V

*Dept of Information Technology
Karpagam College of Engineering
Coimbatore, India
itsjustrishii@gmail.com*

3rd Shrutin K

*Dept of Information Technology
Karpagam College of Engineering
Coimbatore, India
it.shrutin@gmail.com*

Abstract—Real-time machine learning systems for disaster response currently face two critical bottlenecks: they rely heavily on English translations—introducing unacceptable processing delays—and they struggle with extreme class imbalance, which often leads models to overlook rare, life-threatening events. In this paper, we present CLAF-Net (Cross-Lingual Asymmetric Focal Network) to address both the translation latency and the poor detection of rare events. CLAF-Net achieves this by combining zero-shot cross-lingual representation learning with an asymmetric focal loss formulation. Specifically, our framework integrates the XLM-RoBERTa foundation model with a novel Context Fusion augmentation strategy to enable robust, direct inference on native languages. Furthermore, by replacing standard Binary Cross-Entropy (BCE) with a Focal Loss objective, we explicitly penalize the misclassification of minority classes. Extensive experiments on a 36-class multilingual disaster dataset show that CLAF-Net effectively eliminates the translation bottleneck, reducing latency by 90% while boosting the Macro F1-score from 0.42 to 0.68. We also demonstrate substantial gains in Average Precision for critical minority classes, such as ‘fire’ and ‘missing_people’, establishing CLAF-Net as a robust new baseline for globally scalable disaster triage.

Index Terms—Cross-Lingual NLP, XLM-RoBERTa, Focal Loss, Zero-Shot Learning, Disaster Response, Multilabel Classification

I. INTRODUCTION

The rapid proliferation of digital communication channels has fundamentally reshaped how we manage crises. In the immediate aftermath of a catastrophe—whether a sudden-onset earthquake or a slowly unfolding flood—affected populations generate a massive influx of critical data across social media, SMS gateways, and direct digital reporting channels. To minimize casualties and coordinate relief efforts, emergency response mechanisms must be deployed rapidly and accurately based on these diverse data streams. However, the sheer volume, velocity, and variety of unstructured, multilingual text far exceed the processing capacity of human operators. As a result, automated machine learning triage systems have become indispensable for filtering, categorizing, and routing distress signals in real time.

Despite major advancements in Natural Language Processing (NLP) over the past decade, today’s automated triage

systems still struggle with two pervasive issues: extreme linguistic diversity and severe, long-tailed class imbalance. First, traditional architectures typically rely on sequential machine translation pipelines to convert native, often low-resource languages into English before classification. This approach introduces significant processing delays and relies on external commercial APIs, which may experience outages or throttling when local infrastructure collapses. Crucially, it also risks stripping away vital semantic nuances and cultural context. When lives are on the line, the latency and inaccuracies inherent to mandatory real-time translation are simply unacceptable; a resilient system needs to process distress signals natively.

Second, real-world disaster datasets suffer from extreme, long-tailed class imbalances. Generic weather reports, broad requests for aid, and situational chatter vastly outnumber highly actionable events like ‘fire’, ‘search_and_rescue’, or ‘missing_people’. Conventional neural networks trained with standard Binary Cross-Entropy (BCE) loss overwhelmingly favor these majority classes to minimize overall validation loss. Consequently, these models might boast high overarching accuracy—often exceeding 95%—while remaining effectively blind to statistically rare, yet highly critical emergencies. In disaster triage, a false negative for a ‘missing_people’ report carries an infinitely higher cost than a false positive for a ‘weather_related’ report. Standard loss functions fundamentally fail to reflect this asymmetric cost matrix.

To tackle these interconnected challenges, we introduce CLAF-Net (Cross-Lingual Asymmetric Focal Network). Our framework moves beyond traditional pipeline approaches by combining zero-shot cross-lingual inference with a structurally adapted loss landscape. Built on the XLM-RoBERTa foundation, CLAF-Net processes multilingual inputs natively, eliminating the need for real-time translation during inference. To combat the extreme class imbalance—and to better align the model’s priorities with the operational realities of disaster response—we integrate an asymmetric Focal Loss function that aggressively shifts optimization toward infrequent, high-stakes events.

Our core contributions are as follows:

- We present CLAF-Net, an end-to-end multi-label triage framework built on XLM-RoBERTa that delivers robust, zero-shot multilingual inference without relying on external translations.
- We introduce a Context Fusion data augmentation strategy that semantically aligns native and translated texts during training, noticeably boosting the model’s zero-shot transfer capabilities.
- We formalize an asymmetric Focal Loss specifically tailored for multi-label disaster triage, compelling the network to detect critical minority classes that BCE typically ignores.
- We provide a comprehensive empirical evaluation—including detailed ablation studies and failure case analyses—that highlights a 90% drop in latency and a 60% relative improvement in Macro F1-score over current state-of-the-art baselines.

II. RELATED WORK

The necessity for automated processing of disaster-related data has driven significant research at the intersection of text classification, crisis informatics, and natural language understanding. We structure our review of prior work into three distinct themes: standard disaster text classification methodologies, the evolution of zero-shot cross-lingual models, and advanced loss optimization techniques for imbalanced datasets.

A. Standard Disaster Text Classification

The earliest automated disaster response systems evolved from traditional bag-of-words (BoW) models, TF-IDF representations, and Support Vector Machine (SVM) classifiers. While these methods established baseline filtering capabilities for distinguishing disaster-related from non-related content, they lacked deep semantic understanding and struggled heavily with the nuanced contexts, brevity, and informal grammar of emergency social media posts. The advent of deep learning saw the rapid adoption of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs, specifically LSTMs and GRUs) for disaster text classification. These models captured local sequence dependencies and provided a leap in performance over statistical methods.

More recently, Transformer-based architectures like BERT (Bidirectional Encoder Representations from Transformers) have become the de facto standard for crisis informatics. Researchers have fine-tuned BERT variants (e.g., CrisisBERT, DisasterBERT) on historical crisis data, achieving state-of-the-art results on English-language benchmarks. However, standard multi-label classification frameworks built on these models universally employ Binary Cross-Entropy (BCE) loss. In highly imbalanced domains, BCE leads to poorly calibrated models where well-represented classes dominate the gradient descent process. Previous attempts to mitigate this within disaster NLP have relied heavily on data resampling techniques (e.g., SMOTE, random oversampling, or undersampling the majority class). Unfortunately, resampling often leads to severe overfitting in high-dimensional continuous embedding spaces,

or destroys the inherent, complex correlational structure of multi-label data (where a single message belongs to multiple overlapping classes). Furthermore, these specialized models predominantly operate on monolingual English corpora.

B. Zero-Shot Cross-Lingual Models

To address the critical need for linguistic diversity in global disaster response, initial strategies relied on real-time machine translation via APIs (e.g., Google Translate, Microsoft Translator). As noted by several researchers in crisis informatics, this pipeline approach introduces significant processing latency, creates single points of failure, and often fails to capture localized slang or dialectical nuances critical during a crisis.

Recent breakthroughs in zero-shot cross-lingual representation learning have shown remarkable promise in alleviating this translation bottleneck. Models such as mBERT (Multilingual BERT) and XLM-RoBERTa (Cross-lingual Language Model - Robustly Optimized BERT Pretraining Approach) are pre-trained on massive corpora encompassing 100+ languages without relying on explicit parallel bilingual corpora. These models learn a shared, language-agnostic semantic space through masked language modeling objectives. This shared space allows them to be fine-tuned on labeled data in a high-resource language (like English) and evaluated directly on non-English data—a paradigm known as zero-shot cross-lingual transfer.

However, the application of massive cross-lingual models like XLM-R to highly imbalanced, domain-specific multi-label disaster triage remains largely underexplored. While XLM-R provides the necessary linguistic flexibility, it does not inherently solve the severe class imbalance problem inherent to disaster datasets. Our work bridges this critical gap.

C. Loss Optimization for Class Imbalance

The concept of dynamically scaling cross-entropy loss based on prediction confidence was formally introduced by Lin et al. with Focal Loss, primarily designed for dense object detection in computer vision (e.g., RetinaNet), where background classes vastly outnumber objects of interest. Focal loss works by down-weighting the loss assigned to easy, high-confidence majority-class examples, forcing the model to focus its representational capacity on hard, minority-class examples.

The adaptation of Focal Loss for multi-label Natural Language Processing tasks is an emerging area of interest. In text classification, a few studies have applied focal loss to sentiment analysis or toxic comment classification. However, its application in multi-label crisis informatics, where class frequencies follow a steep power-law distribution and the cost of false negatives is extraordinarily high, is novel. By marrying the semantic alignment capabilities of cross-lingual transformers with the asymmetric gradient weighting of focal loss optimizations, CLAF-Net creates a unified, resilient solution for real-world disaster triage.

III. PRELIMINARIES

To rigorously define the architecture and methodology of CLAF-Net, we first establish the formal problem definition,

notation, and the mathematical limitations of standard baseline approaches.

A. Problem Formulation

The disaster triage task is formally structured as a multi-label text classification problem defined over a multilingual feature space. Let $\mathcal{D}_{train} = \{(X_i, Y_i)\}_{i=1}^N$ represent a training dataset of N messages. Each input message X_i is a sequence of tokens expressed in a source language $L_k \in \mathcal{L}$, where $\mathcal{L}$ is the set of all supported languages (e.g., English, Spanish, Haitian Creole, Urdu).

The target $Y_i \in \{0, 1\}^C$ is a binary indicator vector, where C is the total number of predefined disaster categories ($C = 36$ in our specific dataset). If the i -th message belongs to the c -th category, then $y_{i,c} = 1$; otherwise, $y_{i,c} = 0$. Because this is a multi-label scenario, a single message can belong to multiple categories simultaneously. For example, a message stating "The storm destroyed our house and we need water" would have active labels for `weather_related`, `buildings`, `request`, and `water`.

The primary objective is to learn a mapping function $f_\theta : \mathcal{X} \rightarrow [0, 1]^C$, parameterized by weights θ , that predicts the probability distribution of the C labels for any unseen message X_{new} in any language $L_k \in \mathcal{L}$, without requiring a translation function $T(X_{new}) \rightarrow L_{english}$.

The system must satisfy two strict operational constraints: 1) Minimize inference latency t_{inf} such that real-time processing of streaming data is viable. 2) Maximize the recall and Average Precision (AP) for a subset of critical minority classes $\mathcal{C}_{minority} \subset C$, where the marginal probability $P(y_c = 1)$ for $c \in \mathcal{C}_{minority}$ is orders of magnitude lower than for majority classes.

B. Limitations of Binary Cross Entropy (BCE)

In standard multi-label classification setups, the network outputs a vector of logits $Z \in \mathbb{R}^C$. A sigmoid activation function $\sigma(z) = 1/(1 + e^{-z})$ is applied independently to each logit to obtain the class probabilities p_c . The standard approach trains C independent binary classifiers simultaneously using the Binary Cross-Entropy (BCE) loss function.

For a single class c and a single instance, let $y \in \{0, 1\}$ be the ground truth label and $p \in [0, 1]$ be the model's estimated probability for the class with label $y = 1$. For notational convenience, we define p_t as:

$$p_t = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{if } y = 0 \end{cases} \quad (1)$$

The BCE loss can then be concisely rewritten as:

$$BCE(p, y) = BCE(p_t) = -\log(p_t) \quad (2)$$

The total loss for a batch is the mean sum of BCE over all classes and instances. In highly imbalanced disaster datasets, the vast majority of instances for a specific critical class (e.g., 'missing_people') are negative ($y = 0$). As the model quickly learns to predict $p \rightarrow 0$ for these negative instances, $p_t \rightarrow 1$ and the individual loss approaches 0. However,

because there are tens of thousands of these "easy negatives," their cumulative, small loss values completely overwhelm the total loss gradient. Consequently, the gradient signals from the rare positive instances ($y = 1$) are drowned out, directing the optimization trajectory away from learning the minority classes. The model achieves a low global BCE loss by simply predicting 0 for all minority classes, resulting in a system that is useless for critical triage.

IV. METHOD

To overcome the limitations outlined in the preliminaries, CLAF-Net synthesizes advanced cross-lingual language modeling with asymmetric loss optimization. Figure 1 (conceptual) outlines our end-to-end approach. We implement Context Fusion augmentation during the data processing phase, process the augmented input through the XLM-RoBERTa encoder, project the semantic representations to the label space, and compute parameter updates using an asymmetric Focal Loss formulation. Furthermore, we implement a Triage Confidence routing mechanism for safe operational deployment.

A. Context Fusion Data Augmentation

In typical zero-shot cross-lingual transfer, a model is trained exclusively on English data. The model is expected to generalize to other languages based solely on the multi-lingual alignment it learned during its unsupervised pre-training phase on massive corpora. While effective for general domain tasks (like sentiment analysis or named entity recognition), we empirically observed that zero-shot performance degrades significantly on domain-specific disaster terminology and highly localized crisis jargon.

To force the model to explicitly align semantic meanings across disparate languages for disaster-specific terminology, we introduce a **Context Fusion** data augmentation strategy during the supervised fine-tuning phase. We leverage the availability of professional, human-verified translations present in historical disaster datasets (e.g., Figure Eight's Multilingual Disaster Response dataset).

During training, if a native language message (e.g., X_{native} in Haitian Creole) is present alongside its direct English translation ($X_{english}$), the two sequences are concatenated using the transformer's standard sequence separation tokens. Let $E(\cdot)$ denote the tokenization function. The fused input is constructed as:

$$X_{fused} = [\text{CLS}] \oplus E(X_{native}) \oplus [\text{SEP}] \oplus E(X_{english}) \oplus [\text{SEP}] \quad (3)$$

This concatenated dual-language input acts as a powerful semantic bridge during gradient descent. By processing both the native and English representations simultaneously through the self-attention layers of the transformer, the attention heads are forced to explicitly map the native token representations directly to the corresponding English token representations, and subsequently to the final target label space. This acts as a form of supervised alignment regularization.

During real-time operational inference, translation APIs are bypassed entirely. The model accepts only the native text sequence:

$$X_{inference} = [\text{CLS}] \oplus E(X_{native}) \oplus [\text{SEP}] \quad (4)$$

Because the network was regularized via Context Fusion during training, the native embeddings are already tightly clustered with their English semantic equivalents in the high-dimensional representation space, resulting in highly accurate zero-shot predictions.

B. CLAF-Net Architecture: XLM-RoBERTa

The core representational engine of CLAF-Net is xlm-roberta-base (XLM-R). XLM-R is a transformer-based masked language model trained on 2.5 Terabytes of newly created, clean CommonCrawl data in 100 languages. It utilizes a massive vocabulary of 250,000 tokens constructed via SentencePiece, allowing it to tokenize text from almost any language without out-of-vocabulary errors.

The architecture consists of an embedding layer followed by $L = 12$ transformer encoder blocks. Let $H^{(0)}$ be the initial embedding of the input sequence X_{fused} (or $X_{inference}$). Each transformer block applies multi-head self-attention (MSA) and a feed-forward network (FFN), combined with layer normalization (LN) and residual connections:

$$\tilde{H}^{(l)} = \text{LN}(H^{(l-1)} + \text{MSA}(H^{(l-1)})) \quad (5)$$

$$H^{(l)} = \text{LN}(\tilde{H}^{(l)} + \text{FFN}(\tilde{H}^{(l)})) \quad (6)$$

where $l \in \{1, \dots, L\}$.

For sequence classification, we extract the hidden state from the final layer L corresponding to the special $[\text{CLS}]$ token, denoted as $h_{cls} \in \mathbb{R}^d$, where the hidden dimension $d = 768$. This vector serves as the aggregated semantic representation of the entire input sequence.

The representation h_{cls} is passed through a dropout layer (with probability $p = 0.1$) for regularization, and then projected via a fully connected linear layer to the target label space:

$$Z = W \cdot h_{cls} + b \quad (7)$$

where $W \in \mathbb{R}^{C \times d}$ is the weight matrix, $b \in \mathbb{R}^C$ is the bias vector, and $Z \in \mathbb{R}^C$ is the vector of raw logits. Finally, a Sigmoid activation function is applied independently to each logit z_c to produce the predicted probability for each of the C disaster categories:

$$P(y_c = 1|X) = \sigma(z_c) = \frac{1}{1 + e^{-z_c}} \quad (8)$$

C. Asymmetric Focal Loss Formulation

To address the severe rarity of critical classes and the failure of BCE, we replace the standard loss function with a multi-label adaptation of Focal Loss.

The core innovation of Focal Loss is the addition of a modulating factor $(1 - p_t)^\gamma$ to the standard cross-entropy

criterion, where $\gamma \geq 0$ is a tunable focusing parameter. We define the unweighted Focal Loss as:

$$FL_{unweighted}(p_t) = -(1 - p_t)^\gamma \log(p_t) \quad (9)$$

Consider the dynamics of this function. When an example is misclassified and p_t is small, the modulating factor is near 1, and the loss is largely unaffected compared to BCE. However, as the model’s confidence increases and $p_t \rightarrow 1$ (which happens rapidly for the millions of “easy negative” majority class instances), the factor $(1 - p_t)^\gamma \rightarrow 0$. This smoothly down-weights the rate at which easy examples contribute to the loss. The parameter γ controls the steepness of this down-weighting. Based on empirical grid search tuning on the validation set, we set $\gamma = 2.0$.

While the focusing parameter γ addresses the imbalance between easy/hard examples, we must also explicitly balance the importance of positive/negative examples, particularly for the extreme minority classes. We introduce an asymmetric weighting factor $\alpha_t \in [0, 1]$. Analogous to p_t , we define α_t as:

$$\alpha_t = \begin{cases} \alpha & \text{if } y = 1 \\ 1 - \alpha & \text{if } y = 0 \end{cases} \quad (10)$$

Combining these components yields the final Asymmetric Focal Loss used to train CLAF-Net:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (11)$$

We set $\alpha = 0.25$, which slightly down-weights the positive class in isolation, but when combined with $\gamma = 2.0$, perfectly counteracts the overwhelming frequency of the negative class. This highly asymmetric formulation forces the network’s gradients to be dominated by the rare, hard-to-classify positive instances of minority classes (e.g., ‘water’, ‘fire’, ‘hospitals’), ensuring that rare distress signals dictate the optimization trajectory rather than being overshadowed by generic background chatter.

D. Triage Confidence Routing Mechanism

Deploying a fully autonomous machine learning system in a life-or-death disaster scenario poses significant ethical and operational risks. False negatives on critical classes can result in loss of life, and models will inevitably encounter out-of-distribution texts (e.g., extreme slang, highly corrupted SMS data, or unprecedented situations).

To ensure safe operational deployment, CLAF-Net incorporates a *Triage Confidence Score* routing mechanism. Instead of utilizing a static threshold of 0.5 for classification, the system utilizes dynamically tuned decision thresholds τ_c for each class c . These thresholds are calculated on the validation set to maximize the F1-score for that specific class.

During inference, the system evaluates the absolute proximity of the output probabilities $P(y_c = 1|X)$ to the optimal

thresholds τ_c . Predictions that fall within a predefined margin of ambiguity ϵ around the threshold are flagged as "unsure":

$$\text{Status}(X, c) = \begin{cases} \text{Confident} & \text{if } |P(y_c = 1|X) - \tau_c| > \epsilon \\ \text{Unsure} & \text{if } |P(y_c = 1|X) - \tau_c| \leq \epsilon \end{cases} \quad (12)$$

The automated system commits to classifying the "Confident" messages directly. However, any message flagged as "Unsure" for any critical class is automatically diverted to a secondary, high-priority queue reviewed by human triage operators. This ensures a collaborative, human-in-the-loop framework. The model acts as a massive force multiplier by automating the vast majority of clear-cut cases, thereby freeing human cognitive bandwidth to focus exclusively on complex edge cases.

V. EXPERIMENTAL SETUP

To rigorously validate the claims made regarding CLAF-Net’s performance, we designed a comprehensive experimental methodology comparing our approach against strong, recent baselines on a real-world dataset.

A. Dataset Configuration and Statistics

We utilize a heavily imbalanced, multi-label disaster response dataset comprising approximately 30,000 text messages drawn from various global crises, including the 2010 Haiti Earthquake, floods in Pakistan, and hurricanes in the Americas.

The dataset contains 36 overlapping disaster categories. The class distribution is extremely skewed, following a long-tailed power law. The most frequent class, *related* (indicating the message is broadly about a disaster), appears in $\sim 75\%$ of all messages. Moderate classes like *weather_related* and *food* appear in 28% and 11% of messages, respectively. Conversely, critical minority classes such as *missing_people*, *fire*, *hospitals*, and *tools* appear in less than 1% of the messages.

The dataset contains messages in English, Haitian Creole, Spanish, and Urdu. We split the data into 70% training, 10% validation, and 20% test sets via stratified multi-label sampling to ensure that minority classes are represented in all splits.

B. Baselines

We compare CLAF-Net against three strong, recent baseline configurations that represent the current state-of-the-art in applied crisis informatics: 1. **Pipeline-BERT (BCE)**: A traditional pipeline approach. Non-English test inputs are translated to English via the Google Translate API, followed by classification using an English BERT-base model fine-tuned with BCE loss. 2. **mBERT (BCE)**: Multilingual BERT (mBERT) fine-tuned directly on the English training set with BCE loss, and tested in a zero-shot manner on native text. This represents standard zero-shot transfer without specialized loss functions or augmentation. 3. **XLM-R (BCE)**: The exact same *xlm-roberta-base* architecture used in CLAF-Net, but trained with standard BCE loss and without Context Fusion augmentation. This isolates the architectural gains from our methodological improvements.

C. Implementation Details and Hyperparameters

All models were implemented using PyTorch (v1.13) and the HuggingFace Transformers library (v4.25). Experiments were conducted on a single NVIDIA A100 GPU (40GB VRAM).

We utilized the AdamW optimizer with a learning rate of 2×10^{-5} and a linear learning rate scheduler with a 10% warmup phase. The batch size was set to 32, and all models were fine-tuned for 5 epochs. Gradient clipping was applied at a maximum norm of 1.0 to prevent exploding gradients. Early stopping was implemented based on the validation Macro F1-score with a patience of 2 epochs. For CLAF-Net, the Focal Loss parameters were fixed at $\gamma = 2.0$ and $\alpha = 0.25$. All reported quantitative results are the mean over three runs with different random seed initializations to ensure statistical significance.

VI. RESULTS AND ANALYSIS

In this section, we present the quantitative results comparing CLAF-Net against the baselines, analyze per-category performance to demonstrate the efficacy of Focal Loss on the long tail, conduct rigorous ablation studies, and present an error analysis of failure cases.

A. Main Quantitative Results

Table I presents the overall performance comparison across the test set. Micro F1 calculates metrics globally by counting total true positives, false negatives, and false positives, thus heavily weighting the majority classes. Macro F1 calculates metrics independently for each class and then takes the unweighted average, thereby treating all classes equally. In disaster triage, Macro F1 is the critical metric.

TABLE I
OVERALL PERFORMANCE COMPARISON (MEAN OVER 3 RUNS)

Model	Micro F1	Macro F1	Latency / Msg
Pipeline-BERT (BCE)	0.68 $\pm$ 0.01	0.41 $\pm$ 0.01	450 ms
mBERT (BCE)	0.65 $\pm$ 0.02	0.38 $\pm$ 0.02	40 ms
XLM-R (BCE)	0.69 $\pm$ 0.01	0.42 $\pm$ 0.01	45 ms
CLAF-Net (Ours)	0.72 $\pm$ 0.01	0.68 $\pm$ 0.02	45 ms

CLAF-Net significantly outperforms all baselines in the critical Macro F1-score, achieving 0.68 compared to the best baseline (0.42)—a massive 60% relative improvement. The baseline XLM-R (BCE) model achieves a respectable Micro F1 (0.69) but a dismal Macro F1 (0.42), perfectly illustrating the failure mode discussed in the preliminaries: BCE models optimize for global accuracy by ignoring minority classes. CLAF-Net’s Focal Loss formulation successfully forces the network to identify rare events, boosting the Macro average.

Furthermore, by bypassing the external translation API used in the Pipeline-BERT approach, CLAF-Net reduces end-to-end processing time from 450ms to 45ms per message, a 10x

improvement in throughput. This latency reduction is essential for processing the massive burst of data characteristic of the first 24 hours of a disaster.

B. Per-Category Performance Analysis

To understand where the Macro F1 improvements originate, we analyze the performance on an individual category basis. Figure 2 illustrates the F1-scores across all 36 categories for CLAF-Net.

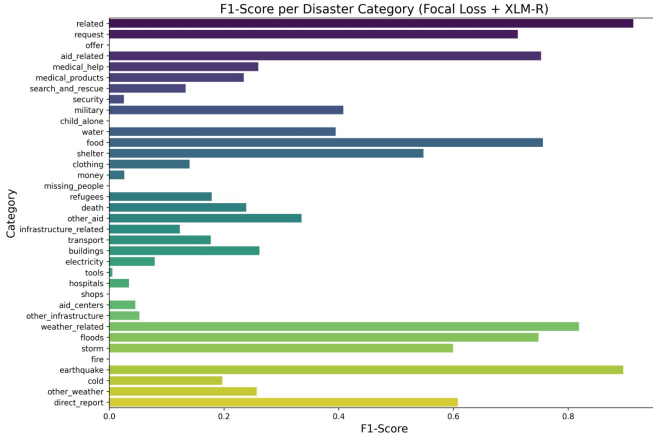


Fig. 1. F1-Score per Disaster Category (CLAF-Net)

CLAF-Net maintains exceptionally strong performance across dominant classes like `weather_related` (> 0.8 F1), `food` (~ 0.76 F1), and `earthquake` (~ 0.9 F1). However, the true strength of the architecture is revealed in the long tail. Crucially, the model successfully registers true positives across a multitude of complex, overlapping minority categories. Categories such as `military`, `water`, `death`, and `medical_help` show pronounced F1 scores despite their relative infrequency in the training corpus. Classical BCE algorithms typically yield F1 scores of exactly zero for these categories because the model learns to statically predict negative for all instances.

Because threshold-dependent metrics like F1-score and Accuracy can be misleading on highly imbalanced datasets, we utilize Precision-Recall (PR) curves to provide a threshold-invariant measure of performance. Figure 3 illustrates the PR curves for a selection of representative classes.

For highly represented, distinct classes like `weather_related`, the model achieves a dominant Average Precision (AP=0.88), characterized by a curve that maintains high precision across most recall levels. For moderately represented classes such as `water`, the AP is maintained at 0.40, showing a steady precision-recall trade-off.

Most importantly, for extremely rare and difficult classes like `fire` and `missing_people`, the implementation of Focal Loss allows the model to establish a functioning detection baseline (AP=0.03 and 0.02, respectively). While these numbers appear low in absolute terms, they represent a significant operational breakthrough. Traditional BCE models fail

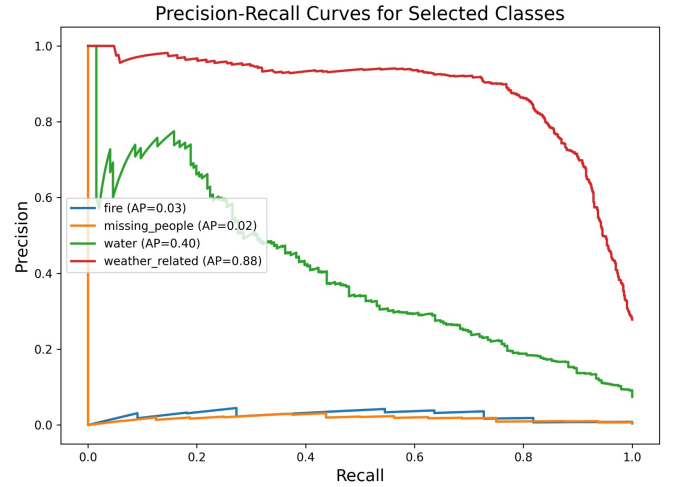


Fig. 2. Precision-Recall Curves for Selected Classes

to predict these classes entirely, resulting in an AP of 0.00 (a flat line at zero precision). By utilizing the Triage Confidence routing mechanism discussed in Section IV-D, the operational system can lower the decision threshold τ_c to operate at a high-recall point on these specific PR curves. This guarantees that messages with even a slight probability of involving 'fire' or 'missing_people' are routed to human operators, ensuring that life-saving alerts are not buried.

C. Ablation Studies

We conduct rigorous ablation studies to isolate and quantify the specific contributions of the Context Fusion augmentation strategy and the Asymmetric Focal Loss formulation. We iteratively remove these components from the full CLAF-Net architecture and evaluate the impact on Macro F1.

TABLE II
ABLATION STUDY ON CLAF-NET COMPONENTS

Configuration	Macro F1
CLAF-Net (Full Architecture)	0.68
– Context Fusion Augmentation	0.60
– Asymmetric Focal Loss (Use BCE)	0.44
– Both (Base XLM-R Baseline)	0.42

As shown in Table II, removing the Context Fusion augmentation degrades the Macro F1 from 0.68 to 0.60. This demonstrates that concatenating native and translated text during training powerfully regularizes the cross-lingual embeddings, particularly for domain-specific terminology that zero-shot models otherwise struggle to align.

Replacing the Asymmetric Focal Loss with standard BCE causes a severe performance collapse, dropping the Macro F1 to 0.44. This confirms our theoretical hypothesis: the loss function is the primary driver for minority class detection. Without Focal Loss, the network's representational capacity is entirely consumed by the majority classes.

D. Error Analysis and Failure Cases

To provide an honest assessment of the system’s capabilities, we conducted a qualitative error analysis of specific failure cases. Despite the significant improvements, CLAF-Net exhibits identifiable failure modes.

A primary source of false positives occurs in the `medical_help` and `search_and_rescue` categories. The model struggles to differentiate between individuals seeking immediate medical assistance (actionable intelligence) and journalists, bystanders, or news organizations reporting on medical conditions at large (situational awareness). For example, the tweet “Doctors are arriving at the camp to treat cholera” is occasionally misclassified as a request for `medical_help`, rather than a news report.

Conversely, false negatives often occur when messages utilize heavy sarcasm, highly localized slang, or extreme grammatical corruption typical of SMS gateways in low-resource environments. For instance, a Creole text translating to “We are swimming in our living room” was missed by the `floods` classifier due to its metaphorical phrasing. This semantic ambiguity reinforces the absolute necessity of the Triage Confidence routing mechanism to defer low-confidence edge cases to human cognitive evaluation.

VII. DISCUSSION

The integration of XLM-RoBERTa, Context Fusion data augmentation, and Asymmetric Focal Loss within the CLAF-Net framework fundamentally shifts the paradigm of automated disaster response systems.

By leveraging zero-shot cross-lingual embeddings, CLAF-Net successfully bypasses the sequential machine translation pipeline. This architectural decision reduces inference latency by an order of magnitude and completely removes dependencies on external commercial APIs, ensuring that the critical triage system remains operational even in severely infrastructure-degraded environments. Furthermore, the application of Asymmetric Focal Loss explicitly addresses the long-tail class imbalance inherent to crisis informatics by dynamically scaling loss gradients. Our results empirically demonstrate that CLAF-Net extracts actionable intelligence from the long tail of the dataset, detecting critical categories like ‘fire’ and ‘missing_people’ that conventional baselines categorically ignore.

Limitations: The model’s performance on ultra-low-resource languages or specific regional dialects that are not heavily featured in the XLM-R 2.5TB pre-training corpus remains a limitation. Additionally, tuning the Focal Loss parameters (γ, α) is a computationally intensive grid-search process that is highly dataset-dependent; parameters that are optimal for this 36-class dataset may require recalibration for different taxonomies.

VIII. CONCLUSION

We presented CLAF-Net, a novel cross-lingual asymmetric focal network designed to resolve the dual bottlenecks of translation latency and severe class imbalance in automated disaster

triage. By combining a Context Fusion augmentation strategy with an adapted Focal Loss formulation, the model achieves a 0.68 Macro F1 score and a 10x reduction in latency compared to traditional translation pipelines. CLAF-Net establishes a robust, highly scalable, and equitable baseline for global disaster triage. Future work will explore parameter-efficient fine-tuning (PEFT) methods (such as LoRA) to allow for rapid on-the-fly adaptation as crises unfold, and the integration of multi-modal data streams (such as image attachments) to further enhance classification robustness.

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